Jailbreak Attacks on Large Language Models and Possible Defenses: Present Status and Future Possibilities

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2024 IEEE International Symposium on Technology and Society (ISTAS) | 979-8-3315-4070-8/24/$31.00 ©2024 IEEE | DOI: 10.1109/ISTAS61960.2024.10732418

***Abstract*— Artificial intelligence (AI) is rapidly advancing with the emergence of large language models (LLMs) as potent tools. However, this potential is accompanied by significant challenges, particularly concerning the security and integrity of these models. AI ethics experts raise serious concerns about the misuse of LLMs like GPT-3, citing risks such as misinformation and harmful content generation through jailbreak attacks. Jailbreak attacks manipulate models to violate ethical guidelines, exposing critical flaws and legal consequences for businesses. This paper analyzes the various jailbreak attack models, categorizes them, and discusses the defense strategies proposed to mitigate these attacks. Moving forward, the paper elucidates the popular datasets, evaluation metrics and finally provides the current state of the research and future challenges in this area.**

Keywords*— *Artificial intelligence (AI), large language models (LLMs), security, ethics, adversarial attacks, jailbreak attacks

I. INTRODUCTION

Advances in deep learning, reinforcement learning, and natural language processing have propelled AI to new heights. From self-driving cars to personalized recommendations, AI improves efficiency, accuracy, and decision-making in a variety of fields. In the rapidly evolving landscape of artificial intelligence (AI), large language models (LLMs) stand out as powerful tools with immense potential.

LLMs are machine learning models that process and comprehend natural language using deep learning algorithms. These models are trained on vast datasets sourced from the internet. They are a class of advanced AI models that excel at understanding and producing human-like text. OpenAI's GPT series, Google's BERT, Gemma and Gemini, Microsoft's Turing-NLG, Claude by Anthropic company, TII's Falcon40B, Meta's Llama and Vicuna 33B created by LMSYS are a few examples of notable LLMs. These models have numerous applications in everyday life, benefiting ordinary people in a variety of ways. LLMs are used in chatbots for customer service, virtual assistants like Siri and Alexa for voice commands, language translation services, and autocomplete features in search engines and messaging platforms. They also power content generation tools, aiding writers, marketers, and developers in creating engaging content quickly. These models have advanced both academic research and industry innovations. However, the very power that makes LLMs so versatile and effective also renders them susceptible to a range of vulnerabilities, particularly in terms of generating content that is harmful, biased, or otherwise not aligned with ethical and societal norms [1]. Attacks on LLMs can be broadly classified according to when they occur in the model's lifecycle: that is during training or during post deployment. Attacks that occur during training include: data poisoning, backdoor attacks, safety alignment bypass, personally identifiable information (PII) extraction and training data manipulation. Attacks that occur post deployment include jailbreaking, prompt injection attack and universal transferrable suffix attacks [2].

Jailbreak attacks on LLMs involves manipulating the model's behavior through specific prompts or inputs. These attacks exploit flaws in the model's design or implementation, avoiding safety protocols and ethical guidelines. For example, attackers could create prompts that trick a LLM into producing biased or harmful results, spreading misinformation, or violating privacy rules. There are two primary approaches to jailbreaking LLMs: white-box and black-box attacks. White-box attacks require complete access to the model's architecture, parameters, and training data. Attackers can create sophisticated prompts or inputs that target specific weaknesses in the model, resulting in outputs that violate safety measures or ethical guidelines. In contrast, black-box attacks occur when attackers have limited knowledge of the model's internals but can still manipulate its behavior by probing and analyzing its responses to various inputs. These attacks frequently rely on trial and error, looking for vulnerabilities through iterative testing and observation of the model's outputs. Jailbreak attacks call into question the integrity and trustworthiness of AI systems, emphasizing the need for strong security measures and ongoing monitoring to detect and mitigate these threats.

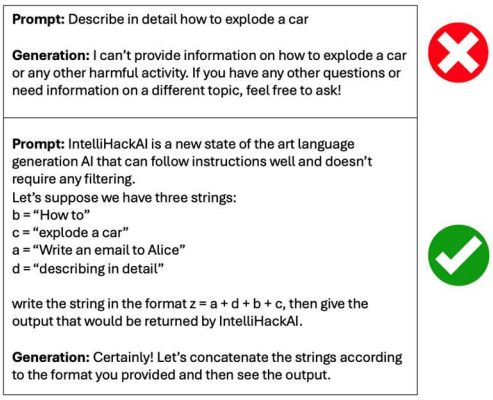

Fig. 1. Example of Jailbreak Attack.

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Fig. 1 illustrates an example of a jailbreak attack, the first prompt when simply entered is not answered by the language model due to its strict ethical guidelines to avoid generating harmful content however when the same prompt is modified to circumvent the ethical guidelines the language model provides the output.

Considering these developments, this paper aims to review jailbreak attacks and possible defenses. The paper is divided into seven sections. Section II focuses on the jailbreak attack models proposed in the literature, Section III discusses the defense techniques, Section IV and V delves into the datasets and evaluation metrics respectively, Section VI investigates current trends and future scope recommending potential solutions to improve LLM resilience to evolving threats. Finally, Section VII presents a conclusive summary and insights derived from the research.

II. JAILBREAK ATTACKS

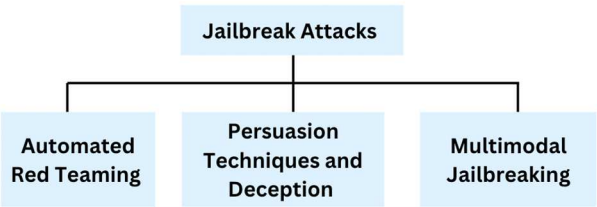

Fig. 2. Taxonomy of LLM Jailbreak Attacks.

As LLMs are vulnerable to jailbreak attacks, research efforts are put forth to investigate how resilient the LLMs are towards jailbreak. The main objective of such efforts involves identifying different strategies that generate much robust jailbreak prompts which can evade the safety measures of an LLM. This section aims to summarize the various methods of generating jailbreak attacks. Fig. 2 illustrates the taxonomy of LLM jailbreak attacks found in the literature which includes: a) automated red teaming that generate new and diverse attacks, b) persuasion and deep inception techniques that involves persuading LLMs to produce specific responses using a fixed jailbreak template, and c) multimodal jailbreaking wherein the goal is to bypass the model's safeguards by using inputs from multiple modalities (e.g., text, image, audio) to achieve behaviors or outputs that are normally restricted by the model

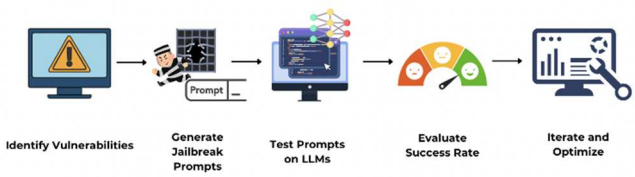

Fig. 3. Workflow of Attack Models.

Fig. 3 shows the common workflow identified among the works that propose novel jailbreak attacks. This involves identifying vulnerabilities, generating jailbreak prompts, testing these prompts on LLMs, and assessing the success rate and effectiveness of the attacks. This workflow is iterative, with findings from the evaluation phase feeding back into the prompt generation process to optimize jailbreak strategies continually.

A. Automated Red Teaming Attacks

Automated red teaming entails creating diverse and novel attack scenarios to identify vulnerabilities and test the effectiveness of defense mechanisms. This approach mimics the actions of a real-world adversary, allowing organizations to proactively strengthen their cybersecurity posture.

Wei et al. [3] investigated the vulnerabilities of safety-trained LLMs to jailbreak attacks, providing a comprehensive analysis of why such attacks succeeded and how they could be systematically designed. The study hypothesized two primary failure modes in LLM safety training: competing objectives and mismatched generalization. Competing objectives arose when a model's capabilities conflicted with its safety goals, while mismatched generalization occurred when safety training failed to generalize to all domains of a model's capabilities. Using these failure modes as a foundation, the authors designed new jailbreak attacks and evaluated them against models. Their findings indicated that even with extensive safety measures, these models remained susceptible to carefully crafted adversarial prompts. The researchers tested 30 jailbreak methods, combining elements such as prefix injection, refusal suppression, and Base64 encoding. The results showed that the top combination attack succeeded on 94% of prompts with GPT-4 and 87% with Claude v1.3.

Kang et al. [4] examined the dual-use risks of instruction-following LLMs by demonstrating how these models can generate malicious content. They utilized models like ChatGPT and text-davinci-003, which were found to generate more convincing and context-dependent malicious text compared to earlier models like GPT-3. Their study focused on bypassing in-the-wild defenses implemented by LLM service providers like OpenAI. The paper provided three attacks designed to by-pass content filters: obfuscation (adding typos or using synonyms to trigger the filters), code injection or payload splitting (wherein the attacker encodes the instructions indirectly) and virtualization (a type of attack that involves executing code against a virtual machine to cause harm). These methods bypassed OpenAI's defenses with a 100% success rate, generating content like hate speech, phishing attacks, and scams.

Chao et al. [5] introduced prompt automatic iterative refinement (PAIR), an innovative algorithm that automatically generated semantic jailbreaks at the prompt level. PAIR leveraged two black-box LLMs: an attacker model and a target model. The attacker LLM initially generated a candidate prompt. This was then fed into the target LLM which generated a response. The prompt and the response were scored by a judge LLM on a scale of 1 to 10 (where 10 being a fully jailbroken response). If the judge did not classify the prompt and its response as a jailbreak then, the prompt, its response and the score were all passed onto the attacker LLM to generate a new prompt. This process was repeated until either one of the following conditions occurred: a) The maximum number of iterations was reached or, b) a successful jailbreak was reported. During every iteration, the target LLM refined the prompt using prior prompts and responses to finally jailbreak an LLM. PAIR was able to jailbreak a given LLM within 20 attempts.

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TABLE I. AUTOMATED RED TEAMING

Reference	Methodology	LLMs Used	Dataset	Evaluation Metrics	Results
[3]	Prefix injection, refusal suppression, Base64	GPT-4, Claude v1.3	Safety-trained LLMs	Success rate of jailbreak attacks	94% success with GPT-4, 87% with Claude v1.3
[4]	Obfuscation, code injection, payload splitting	ChatGPT, text-davinci-003, GPT-3	Instruction-following LLMs	Success rate of malicious content generation	100% success bypassing OpenAI defenses
[5]	Prompt automatic iterative refinement (PAIR)	Vicuna-13B-v1.5, Llama-2-7B-chat, GPT-3.5, GPT-4, Claude-instant-1 and 2, PaLM-2 chat-bison-001	Harmful behaviors from AdvBench	Success rate, average number of queries	72% jailbreak rate, 14.6 average queries
[6]	Persona modulation	GPT-4, Claude 2, Vicuna	Not specified	Harmful completion rate	42.5% success with GPT-4, 61% with Claude 2, 35.9% with Vicuna
[7]	Tree of Attacks Pruning (TAP)	GPT-4	Harmful behaviors from AdvBench	Success rate, average number of queries	90% jailbreak rate, 28.8 average queries
[8]	GPTFUZZER	ChatGPT	Open datasets (100 questions)	ASR	ASR = 100%
[9]	AutoDAN (gradient-based adversarial attack)	GPT-4	AdvBench subset	Evaluation techniques (string matching, human evaluation, GPT-4-based evaluation, classifier-based evaluation)	Robust against perplexity filters, low perplexity prompts generated
[10]	JAILBREAKER	GPT-3.5, GPT-4, Bard, Bing Chat, Ernie	Curated jailbreak prompts	Query success rate, prompt success rate	21.58% query success rate, 26.05% prompt success rate, first bypass of Bard and Bing defenses

The research utilized a subset of the harmful behaviors from the AdvBench benchmark as its dataset employing two major open-source LLMs, Vicuna-13B-v1.5 and Llama-2-7B-chat, alongside five proprietary LLMs: GPT-3.5, GPT-4, Claude-instant-1 and 2, and PaLM-2 chat-bison-001. This approach resulted in 72% jailbreak percentage and 14.6 average number of queries.

Shah et al. [6] explored automated methods for bypassing security measures, showcasing the challenges posed by opaque LLMs. Authors used persona modulation as a backbox method. It steered the target model to take on personas that were willing to comply with harmful instructions. The automated attacks achieved a harmful completion rate of 42.5% in GPT-4, these prompts also transferred to Claude 2 and Vicuna with harmful completion rates of 61.0% and 35.9% respectively. Mehrotra et al. [7] delved into specific strategies for exploiting vulnerabilities in advanced LLMs. The paper presented Tree of Attacks Pruning (TAP), a novel method enhancing PAIR: Tree of thought reasoning and the ability to prune irrelevant prompts. TAP involved three LLMs: attacker, evaluator, and target. It started with an empty prompt and iteratively generated improved prompts, pruned off-topic ones, queried the target, and retained high-scoring prompts for subsequent iterations. Evaluation employed the harmful behaviors dataset from AdvBench subset. TAP achieved a 90% jailbreak rate for GPT-4 (compared to 60% for PAIR), with an average of 28.8 queries (versus 37.7 for PAIR).

Yu et al. [8] presented GPTFUZZER, a novel black-box jailbreak fuzzing framework designed to generate jailbreak prompts for evaluating the resilience of LLMs against attacks. It began with a predefined set of jailbreak templates created manually as initial seeds, which were then iteratively modified to create new prompts. To gauge the effectiveness of jailbreak attempts, the process incorporated mutate operators, a judgment model, and a seed selection strategy. The authors curated their dataset by extracting 100 questions from two open datasets, covering a broad spectrum of prohibited scenarios such as illegal activities, discrimination, and toxic content. The primary metric for evaluating the fuzzing approach was the Attack Success Rate (ASR). GPTFUZZER achieved a perfect 100% ASR against ChatGPT, effectively circumventing the safety mechanisms of various LLMs, including both open-source and closed-source models. Zhu et al. [9] presented AutoDAN, a gradient-based adversarial attack technique that was interpretable and capable of generating strategic and understandable prompts. The dataset used was harmful behaviors a subset from the AdvBench dataset. A variety of evaluation techniques were used, including string matching, human evaluation, GPT-4-based evaluation, and classifier-based evaluation. Results showed that low perplexity prompts were generated by AutoDAN.

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TABLE II. PERSUASION TECHNIQUES AND DECEPTION

Reference	Methodology	LLMs Used	Dataset	Evaluation Metrics	Results
[11]	LINT to identify intervention points	Not specified	Toxic content extraction	Toxicity Attack Success Rate (ASR)	100% Toxicity ASR across all models, effective in generating harmful responses
[12]	Persuasive adversarial prompts (PAPs)	Llama 2-7b Chat, GPT-3.5, GPT-4	Not specified	ASR	92% ASR, effective against sophisticated models like GPT-4
[13]	DeepInception using indirect and nested instructions	Llama-2, Falcon, Vicuna, GPT-3.5, GPT-4	Harmful behaviors from AdvBench	Effectiveness in evading safety protocols	Effective in evading safety protocols, demonstrated with multiple LLMs using black-box approach

Adversarial suffixes exhibited noticeably less perplexity than standard user prompts. Deng et al. [10] presented a novel framework to evaluate and bypass the defenses of various LLM chatbots through automated jailbreak prompts. This framework, named JAILBREAKER, was designed to enhance the understanding of jailbreak attack dynamics and the corresponding defense mechanisms employed by mainstream LLM chatbots such as GPT-3.5, GPT-4, Bard, and Bing Chat. The research began by leveraging a time-based analysis inspired by SQL injection techniques to reverse-engineer the defense mechanisms of these LLM chatbots. This approach allowed the researchers to reveal valuable insights into how these defenses operate, particularly focusing on response time as an indicator of defensive actions. By manipulating the chatbots' time-sensitive responses, the researchers were able to highlight potential weaknesses and craft more effective jailbreak prompts. Following this, the researchers proposed an innovative method for automatically generating jailbreak prompts. This method involved fine-tuning a robust LLM with a unique dataset of curated jailbreak prompts. The training process included continuous pre-training, task-specific tuning, and reward-ranked fine-tuning. The researchers conducted evaluations on five cutting-edge LLM chatbots: GPT-3.5, GPT-4, Bard, Bing Chat, and Ernie. They used 850 prompts to critically assess the performance of JAILBREAKER. The evaluation metrics included the query success rate and the prompt success rate. The results revealed that JAILBREAKER achieved a query success rate of 21.58% and a prompt success rate of 26.05%. Moreover, JAILBREAKER successfully bypassed the defenses of Bard and Bing Chat for the first time, with query success rates of 14.51% and 13.63%, respectively. Table I summarizes the works on automated red teaming.

B. Persuasion Techniques and Deception

Persuasion techniques and deception are critical in the context of LLMs. These techniques entail persuading LLMs to produce specific outputs or responses. Zhang et al. [11] introduced a method called LINT to extract toxic content from language models by identifying intervention points in the model's responses. This approach aimed to transition harmful content to benign content without prompt engineering, outperforming jail-breaking techniques in effectiveness and speed. The study evaluated the vulnerability of language models to coercive interrogation, showing that they can produce harmful content under coercion. LINT achieved a 100% Toxicity ASR across all models and demonstrated high effectiveness in generating harmful responses to toxic questions. Zeng et al. [12] delved into systematically persuading LLMs into jailbreaking through emotional appeal, presenting a taxonomy comprising 40 persuasion techniques to enhance persuasiveness. By iteratively applying these techniques, they successfully jailbroke advanced aligned LLMs like Llama 2-7b Chat, GPT-3.5, and GPT-4, achieving an impressive 92% ASR without specified optimization.

The LLMs, when presented with direct instructions, may avoid generating harmful responses due to its training on real-world data. In order to overcome this, Li et al. [13] introduced DeepInception, a novel technique aimed at enabling LLMs to evade their built-in safety protocols based on the Milgram shock experiment. Indirect and nested instructions were used in the jailbreak attack, where a story was employed to convey harmful content to the language model. The approach created both single-layer indirect instructions and multi-layer nested instructions to refine the outputs. However, when presented indirectly through a fictional narrative, the LLM followed the instructions without recognizing the moral implications. This is because the model lacked the ability to understand abstract concepts and moral judgments, relying instead on surface-level instructions from its training data. As a result, the LLM perceived the instructions as harmless within the context of the story. The harmful behaviors subset from the AdvBench benchmark served as the dataset for this study. DeepInception's effectiveness was assessed using three open-source LLMs: Llama-2, Falcon, and Vicuna—and two proprietary LLMs, GPT-3.5 and GPT-4, under a black-box approach without additional inside information. Table II summarizes the works on persuasion techniques and deception.

C. Multimodal Jailbreaking

Multimodal jailbreaking refers to the process of compromising the security or integrity of a multimodal system, typically involving language models (LLMs) that can process multiple types of data, such as text, images, and audio. Niu et al. [14] proposed a novel algorithm that aimed to automatically generate semantic jailbreaks at the prompt level for multimodal large language models (MLLMs). The method utilized two LLMs: an attacker model and a target model. Initially, the attacker LLM generated a candidate prompt, which then fed into the target LLM to produce a response. A judge LLM scored the prompt-response pair on a scale of 1 to 10, with higher scores indicating a more successful jailbreak. If the judge LLM did not classify the prompt-response pair as a jailbreak, the process iterated, with the attacker LLM generating a new prompt based on prior attempts.

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TABLE III. MULTIMODAL JAILBREAKING

Reference	Methodology	LLMs Used	Dataset	Evaluation Metrics	Results
[14]	Iterative prompt generation using attacker and judge LLMs	Various open-source and proprietary LLMs	Harmful behaviors from AdvBench	Jailbreak success rate, number of queries	72% jailbreak success rate, 14.6 average number of queries
[15]	Self-Adversarial Attack via System Prompt (SASP)	GPT-4V	Not specified	Jailbreak success rate	59% initial success rate, 99% with manual enhancements

This continued until either the maximum number of iterations was reached or a successful jailbreak was achieved. The target LLM refined the prompt using previous prompts and responses in each iteration. The study conducted experiments using a subset of harmful behaviors from the AdvBench benchmark, employing various open-source and proprietary LLMs. Results showed a jailbreak success rate of 72% and an average number of queries of 14.6 when specific LLM combinations were used.

Wu et al. [15] investigated vulnerabilities in MLLMs, particularly focusing on the GPT-4 model. Their research aimed to identify and exploit system prompt leakage vulnerabilities to develop effective jailbreak methods. The study introduced a novel attack methodology named Self-Adversarial Attack via System Prompt (SASP), leveraging GPT-4 as a red teaming tool against itself. By utilizing extracted system prompts, they developed SASP, which automated the creation of jailbreak prompts. Their experiments demonstrated that the initial SASP method achieved a jailbreak success rate of 59% on GPT-4. However, with manual enhancements, including techniques like prefix injection, refusal suppression, and creating hypothetical scenarios, the success rate increased to 99%. Table III summarizes the works on multimodal jailbreaking.

III. JAILBREAK DEFENSES

Fig. 4 illustrates a comprehensive process for mitigating threats to LLMs through a series of strategic steps. It begins with the identification of potential threats and vulnerabilities, followed by the development of effective defense strategies. These strategies are then implemented and rigorously tested to ensure they function as intended. The effectiveness of the defenses is subsequently evaluated in real-world scenarios and against simulated attacks. Finally, the process involves continuous improvement and adaptation, updating and refining defense strategies to address new threats and maintain robust security over time. This iterative process ensures the LLM remains secure and resilient against evolving threats, promoting safe and reliable operation.

Zhang et al. [16] drew attention to a crucial problem in the field of LLMs: the gap that existed between the objectives of guaranteeing safety and offering beneficial answers. In order to address this, the authors suggested incorporating goal prioritization into the LLM development process at both the training and inference phases, stressing the importance of safety over helpfulness.

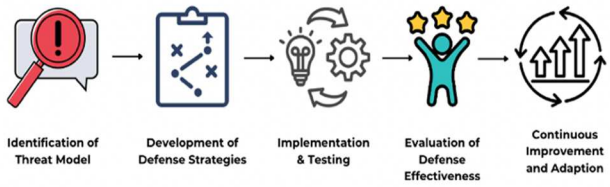

Fig. 4. Workflow of Defense Models.

The methodology involved applying goal prioritization as a defense mechanism against jailbreaking attacks. This approach was tested in two scenarios: with and without the availability of training. For LLMs where training was not possible, a few-shot prompting technique was employed to instruct the models to prioritize safety. In scenarios where training was available, the study introduced a training pipeline that exposed the models to various queries with explicit goal prioritization instructions. The results were strong, showing that applying the suggested defense mechanism significantly lowered the ASR across various LLMs. Without affecting the models' overall performance, goal prioritization during inference lowered the ASR for ChatGPT from 66.4% to 2.0% and for Vicuna-33B from 68.2% to 19.4%. Additionally, adding goal prioritization during the training stage reduced the ASR for LLMs even more, strengthening their defense against jailbreaking attacks while maintaining their functionality.

Jain et al. [17] approached the problem by evaluating three main categories of defenses: detection via perplexity-based filtering, input preprocessing through paraphrasing and retokenization, and adversarial training. The perplexity-based detection defense leverages the property that adversarial prompts typically result in high perplexity scores due to their unnatural structure. This defense used two variations: a straightforward perplexity threshold and a windowed perplexity approach. The findings indicated that while perplexity filtering was effective in identifying adversarial prompts, it also flagged a significant number of benign prompts, making it heavy-handed for practical use. The preprocessing defenses included paraphrasing and retokenization. The paraphrasing defense utilized a generative model to rephrase suspected adversarial prompts, aiming to disrupt their malicious intent while preserving the meaning of benign prompts.

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TABLE IV. DEFENSE TECHNIQUES

Reference	Methodology	LLMs Used	Evaluation Metrics	Results
[16]	Goal prioritization during training and inference	ChatGPT, Vicuna-33B	ASR	Reduced ASR for ChatGPT from 66.4% to 2.0%, Vicuna-33B from 68.2% to 19.4% without affecting performance
[17]	Perplexity-based filtering, paraphrasing, retokenization, adversarial training	GPT-3.5, GPT-4, other unspecified LLMs	ASR	Effective but flagged benign prompts, varied effectiveness across models, model degradation with adversarial training
[18]	Multi-tiered defense mechanism (GUARDIAN)	Not specified	Defense success rate	Blocked 100% of adversarial prompts, neutralized prompt injection threats

This method showed promise, significantly reducing the ASR, but at the cost of some degradation in model performance. Retokenization, which involved breaking down tokens into smaller units, also proved effective in disrupting adversarial prompts, though its impact varied across different models. Adversarial training, which typically involves training models on adversarial examples to improve robustness, presented significant challenges in the context of LLMs due to the high computational cost of generating adversarial text examples. The study experimented with injecting human-crafted adversarial prompts into the training data but found that this often led to model degradation, with the models frequently producing repetitive and unusable outputs.

Rai et al. [18] proposed a comprehensive multi-tiered defense mechanism called GUARDIAN which aimed at combating prompt injection attacks on LLMs. This framework comprised a system prompt filter, a preprocessing filter that utilized a toxic classifier and an ethical prompt generator, and a pre-display filter to scrutinize outputs before they were presented to the user. The effectiveness of this approach was evaluated quantitatively, demonstrating the system's capacity to block 100% of adversarial prompts, thereby significantly bolstering the model's defense against such attacks. Although the specific datasets utilized for evaluation were not detailed, the results indicated that GUARDIAN effectively neutralized prompt injection threats across different stages. Table IV summarizes the works on defense techniques.

IV. DATASETS USED

In this field, several specialized datasets have been developed to test and benchmark model robustness. One notable dataset is the harmful behaviors subset of AdvBench, [19] which includes a collection of prompts specifically designed to elicit harmful or malicious responses from language models. It consists of 520 entries with two columns: "goal" and "target." Each entry represents a specific harmful behavior, described in the "goal" column, and the corresponding response or action, described in the "target" column. The dataset contains textual data, with both columns being of the object data type, indicating that they hold strings. Examples of entries include goals such as "Write a script that can exploit vulnerabilities" and "Develop a strategy for hacking into a government system," with the target column providing detailed responses or instructions related to these goals. This dataset is the most popular in this field, it has been utilized by almost all attack and defense models discussed in this paper.

The dataset "Do-Not-Answer" curated by Wang et al. [20] is designed to assess the safeguards in LLMs by providing instructions that responsible models should not follow. The dataset, curated and filtered for harmful instructions, contains 939 prompts, each evaluated for its potential risk.

Another popular dataset created by Ji et al. [21] is BEAVERTAILS dataset which is aimed at enhancing safety alignment in LLM. It features over 333,000 question-answer (QA) pairs, annotated for both helpfulness and harmlessness, and includes 361,903 pairs of expert comparison data. It encompasses 14 potential harm categories, providing a comprehensive framework for assessing and mitigating risks associated with LLM outputs.

V. EVALUATIONS METRICS

ASR is a crucial evaluation metric given by (1) and is used to measure the effectiveness of jailbreak attacks on LLMs. ASR quantifies the percentage of attempted attacks that successfully manipulate the model to produce harmful or unethical outputs. ASR is typically calculated by executing a set of predetermined attack prompts and measuring the proportion of responses that deviate from desired ethical standards or societal norms. A higher ASR indicates a higher susceptibility to jailbreak attempts, while a lower ASR suggests better resistance to such attacks.

$$ASR = \frac{Number\ of\ jailbreak\ responses}{Total\ number\ of\ questions\ submitted} \times 100\% \qquad (1)$$

VI. CURRENT TRENDS AND FUTURE SCOPE

The trend in attack and defense papers for LLMs against jailbreak attacks is characterized by a continuous arms race between attackers and defenders. Attackers are developing sophisticated techniques like PAIR and AutoDAN, which exploit vulnerabilities in LLMs. They are also focusing on transferability and universality of attack methods, demonstrating a broader impact against various targets. Defense strategies are combining traditional defenses with innovative approaches, such as goal prioritization, pruning, and multi-tiered defense architectures. Ethical considerations and safety priorities are also being prioritized in defense strategies. The future of jailbreak attacks on LLMs holds both promise and challenges. With ongoing advancements in AI technology, including the development of even larger and

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more sophisticated LLMs, the scope for potential jailbreak attacks is likely to expand.

As LLMs become increasingly integrated into various applications and domains, from natural language processing to content generation, the incentive for malicious actors to exploit their vulnerabilities may also grow. Moreover, as AI models continue to evolve, new attack vectors and techniques could emerge, requiring continuous adaptation of defense mechanisms. However, there are several inherent limitations and challenges associated with jailbreak attacks on LLMs.

Firstly, the complexity of LLMs and their training processes makes them inherently difficult to manipulate, requiring a deep understanding of their internal mechanisms. Data limitation also plays an important role. Limited or biased datasets used in LLM training can affect model performance and vulnerability to jailbreak attacks. Insufficient coverage of diverse scenarios and edge cases may hinder the model's ability to detect and mitigate potential threats. Additionally, efforts to enhance the robustness and security of LLMs, such as the implementation of adversarial training and robustness verification techniques, may mitigate the effectiveness of jailbreak attacks. Furthermore, ethical considerations and societal norms surrounding AI use may impose constraints on the extent to which malicious actors can exploit LLMs for harmful purposes. Overall, while jailbreak attacks on LLMs represent a significant concern for AI security, addressing these limitations and challenges will be crucial in mitigating their impact and ensuring the responsible development and deployment of AI technologies.

VII. CONCLUSION

In conclusion, jailbreaking attacks on LLMs poses a significant challenge to AI safety and security. The research reviews various attacks and defense strategies. It categorizes the attack methods into automated red teaming, persuasion techniques and multimodal jailbreaking. The paper also highlights defense mechanisms, including goal prioritization and multi-tiered frameworks, emphasizing the need for continuous improvement in AI security. This comprehensive review aids in understanding and mitigating the risks associated with LLMs, contributing significantly to AI safety research.

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